

# Ch8 Tree-based methods

↳ Introduce a common non-parametric method.

Regression:  $Y = f(x) + \epsilon$

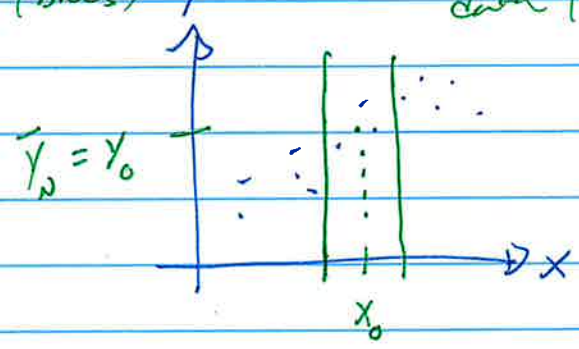
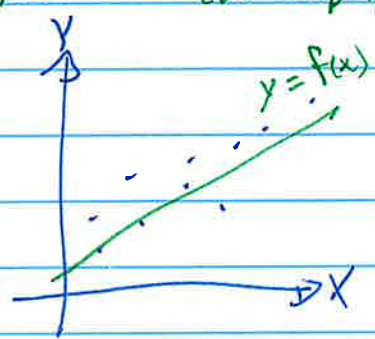
Parametric vs Non-parametric

Linear regression:  $f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$   
Assume  $\beta_i$ 's params to fit

No fixed formula at all  
eg. \*KNN

Pro: if assumption simplifies data  
Con: possible oversimplify (high bias)

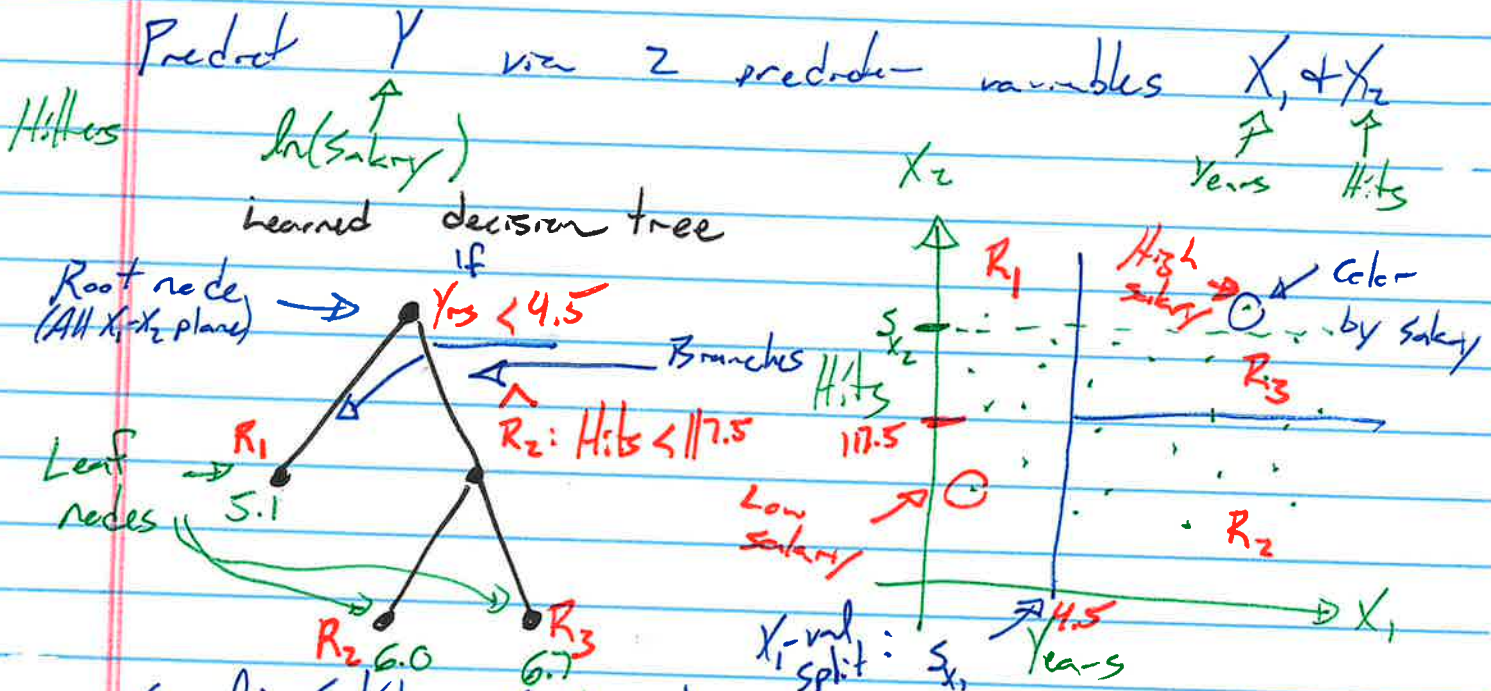
Pro: No assumptions on data  
Con: Sensitive to train data (high var)



2<sup>nd</sup> standard non-parametric approach is decision trees.

Bayes

# Decision tree basics: Regression case.



Goal: Split  $X_1$ - $X_2$  plane in a way which separates  $Y$  as best as possible

How to define? RSS

Iterate until a stop condition is met

Prediction: For new data  $X_0 = \begin{bmatrix} X_{10} \\ X_{20} \end{bmatrix}$ , find which region it belongs to  $R_i$  & predict

$$\hat{Y} = \bar{Y} \text{ for } Y_i \text{ with } X_i \in R_i$$

Avg  $Y$  in same

Tuning

Param: How deep should tree be? <sup>region</sup>

$\hookrightarrow$  Tune via CV.

## Careful version of DT algorithm:

pg 3

$Y$  = resp variable,  $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$  predictor variables.

- ① All data is in one region (root node)
- ② Consider all possible  $2$  region divisions.

$R_1(j, s) = \{X \mid X_j < s\}$  check for all  $s$ -vals as midpoints of sorted  $X_j$ -train data

$R_2(j, s) = \{X \mid X_j \geq s\}$

- ③ Compute residual sum of squares

$$RSS(j, s) = \sum_{i: X_i \in R_1} (y_i - \hat{y}_{R_1})^2 + \sum_{i: X_i \in R_2} (y_i - \hat{y}_{R_2})^2$$

- ④ Choose  $j, s$  which <sup>region  $y_i$  args</sup> minimizes  $RSS(j, s)$  & use as split decision.

- ⑤ Repeat until meet stop condition. 2 options:
  - (i) Min node train population
  - (ii) Split whole way, then prune tree.

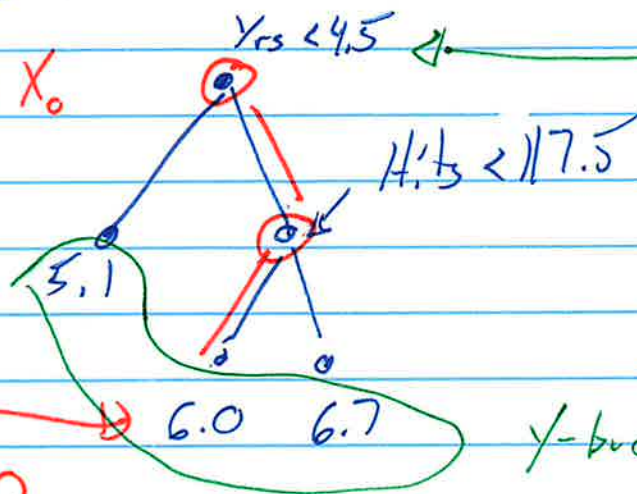
(a) CV

(b) Regularization.

Prediction Trace new data down tree until hit a leaf. Predict leaf node value.

New data:  $X_0$

$Y_{rs} = 6$   
 $Hits = 100$



Balanced over/underfitting: Regularization

# regions of  $X$  (leaf count)

$$R_\alpha(T) = \sum_{m=1}^{|T|} \sum_{i \in R_m} (x_i - \hat{y}_{R_m})^2 + \alpha |T|$$

Annotations for the equation:

- $|T|$ : leaf shrink
- $\sum_{m=1}^{|T|} \sum_{i \in R_m} (x_i - \hat{y}_{R_m})^2$ : RSS per leaf, sum over leaves
- $\alpha |T|$ : Shrinkage of leaf cnt

$\alpha = \text{big}$ , very few leaves (small tree)  
 $\alpha = \text{small}$ , more leaves (big tree)

delete some leaves.

Recall LR regularization:

(Ridge)

$$RSS + \text{Shrinkage} = \sum (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^p \beta_i^2$$

(LASSO)

$$+ \lambda \sum_{i=1}^p |\beta_i|$$

Annotations for the LASSO equation:

- $\sum_{i=1}^p |\beta_i|$ : var. sel & shrink

## Decision tree pros/cons:

Pro

Simple + easy to interpret  
Data driven  
No assumptions on data

Con

Often over simple  
Sensitive to train data

Often lower accuracy than other approaches

To improve accuracy, usually <sup>via bootstrapping</sup> train 100's of trees & combine into a single ensemble model.  
avg predictions

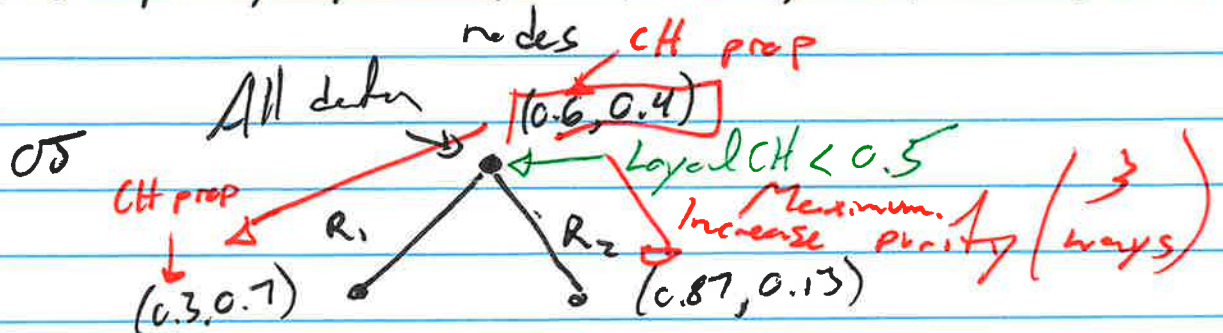
DT basics for regression and.

<sup>Bagging</sup>  
★ Random Forest  
★ Boosting  
top for tabular data

How to extend to classification?

Almost all same ideas as regression tree, but need a new split decision.

Node purity: More pure if very separate classes in nodes



PS 6

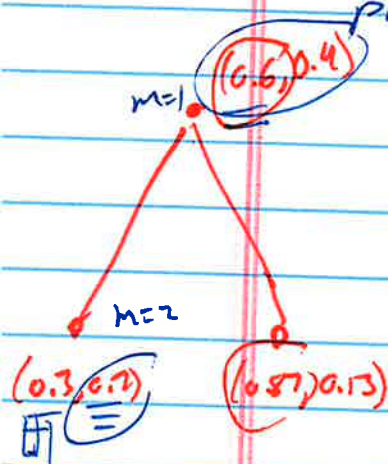
category  
↓

Detailed version Predict qualitative  $Y$  from  $X_1, X_2, \dots, X_p$

$R_j$  = regions as in regression.

Predict:  $\hat{y}(x_0) = \text{majority class in } R_j$   
Best splits if  $x_0 \in R_j$

Generate new regions, measure new node  
 purity 3 ways: min error rate prop of class  $k$  in region  $m$



① Max accuracy of class  $k$ :

$$1 - \max_k \hat{p}_{mk}$$

Ex  $1 - 0.6 = 0.4$   
 $1 - 0.7 = 0.3$

lower, improve accuracy by  $0.4 - 0.3 = 0.1$

Issue: fails for 3+ classes

② Gini index:  $G_m = 1 - \sum \hat{p}^2$

Ex  $G_1 = 1 - 0.6^2 - 0.4^2 = 0.48$  emphasizes large  $\hat{p}$ 's

$G_2 = 1 - 0.3^2 - 0.7^2 = 0.42$  decreases by 0.06

③ Cross entropy: from info theory more relaxed than min error rate

$$L_m = - \sum_k \hat{p}_{mk} \cdot \ln(\hat{p}_{mk}) \quad \text{in } [0, 1],$$

↑  
natural log

over all classes

smaller is better

Still prune a tree same as before but slightly modified.

$$R_{\alpha}(T) = \sum_{m=1}^{|T|} Q_m(T) + \alpha |T|$$

$|T|$  measures avg impurity (MCR, Gini, cross-entropy)  
 $\sum_{m=1}^{|T|} Q_m(T)$  similar to RSS  
 $\alpha |T|$  shrinkage term

choose  $\alpha$  via CV.